

Intro to Rational Expressions – Fractions and Exponents Review – Lesson

MCR3U

Fractions Review

Adding and Subtracting Fractions

a) $\frac{1}{2} + \frac{1}{8}$

b) $\frac{2x}{3y} - \frac{y}{2}$

Always find a common denominator when adding or subtracting fractions!

Multiplying and Dividing Fractions

a) $\frac{3}{2} \cdot \frac{4}{5}$

b) $\frac{2}{3} \div \frac{4}{3}$

You do NOT need a common denominator when multiplying or dividing fractions!

Rule: We can NEVER have a fraction with a denominator of 0. Why?

Rule: Cross multiplication of fractions only happens when...

Rule: We can cancel out ONLY when multiplying fractions

Rule: We can NOT cancel out when adding or subtracting fractions

Name	Rule	Examples
Adding and Subtracting Monomials	COMBINE LIKE TERMS! (do not change common variables and exponents)	$3x^2y + 2x^2y =$
Product Rule	$x^a \cdot x^b =$	$(-2x^2y)(3x^3y^2) =$
Quotient Rule	$\frac{x^a}{x^b} =$	$\frac{28x^5}{42x} =$
Power of a Power Rule	$(x^a)^b =$	$(-2x^3)^2 =$
Negative Exponent Rule	$x^{-a} =$	$\frac{4x^2}{8x^5} =$
Exponent of Zero	x^0	$87^0 =$

Simplify the following rational expressions using exponent laws.

a) $\frac{12k^2m^8}{4k^5m^5}$

b) $\frac{5c^3d \cdot 4c^2d^2}{(2c^2d)^2}$

$$\mathbf{c)} \quad \frac{(3xy)^3}{9x^4y^4}$$

$$\mathbf{d)} \quad \frac{(2z^3)^{-2}}{w^5z^2}$$

$$\mathbf{e)} \quad \frac{(x^{-4})^5x^3}{3x^{-1}}$$

Combining fractions and exponents

$$\mathbf{Ex.} \quad \frac{3x^3}{2x^2} + \frac{4y^4}{3y}$$